

Chemistry - CET Practice Question Paper

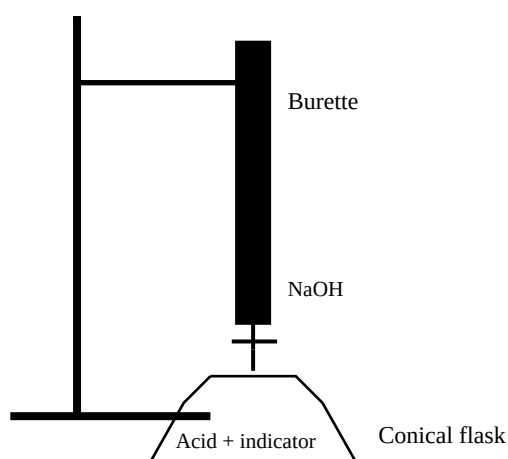
10 Multiple Choice Questions | 4 options each | Includes diagram-based questions

Questions	10
Suggested Time	15 minutes
Marking	Practice set - no negative marking assumed

Q1. Which quantum number primarily determines the shape of an atomic orbital?

- A. Principal quantum number
- B. Azimuthal quantum number
- C. Magnetic quantum number
- D. Spin quantum number

Q2. In the titration setup shown, the solution in the burette is NaOH and the flask contains an acid with indicator. At equivalence for a strong acid-strong base titration, the pH is approximately:



- A. 2
- B. 5
- C. 7
- D. 12

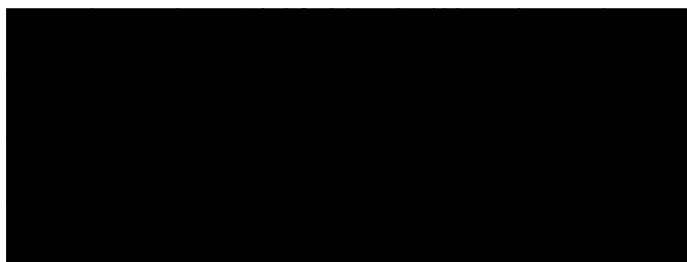
Q3. For the reaction $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$, how many moles of H_2 are required for 2 moles of N_2 ?

- A. 2 mol
- B. 3 mol
- C. 4 mol
- D. 6 mol

Q4. Which substance acts as a reducing agent in $\text{Zn} + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu}$?

- A. Zn
- B. CuSO_4
- C. ZnSO_4
- D. Cu

Q5. In the simplified periodic-table diagram, which labeled position is most likely to represent a noble gas?



- A. A
- B. B
- C. C
- D. None

Q6. The hybridization of carbon in methane (CH_4) is:

- A. sp
- B. sp^2
- C. sp^3
- D. dsp^2

Q7. Which colligative property is most directly used to determine molar mass of macromolecules with high accuracy?

- A. Relative lowering of vapour pressure
- B. Elevation of boiling point
- C. Osmotic pressure
- D. Depression of freezing point

Q8. The IUPAC name of $\text{CH}_3\text{-CH}_2\text{-OH}$ is:

- A. Methanol
- B. Ethanol
- C. Ethanal
- D. Ethanoic acid

Q9. Which gas is evolved when dilute HCl reacts with sodium carbonate?

- A. O₂
- B. H₂
- C. CO₂
- D. Cl₂

Q10. A catalyst increases reaction rate mainly by:

- A. Increasing enthalpy change
- B. Lowering activation energy
- C. Changing equilibrium constant
- D. Increasing product energy